

Parcial 2 - Juan Jerónimo Castaño Rivera

$$x(t) = |A \sin(2\pi f_0 t)|^2 \quad t \in \left[-\frac{1}{2f_0}, \frac{1}{2f_0}\right] \quad A, f_0 \in \mathbb{R}^+$$

$$T = \frac{1}{2f_0} - \left(-\frac{1}{2f_0}\right) = \frac{2}{2f_0} = \frac{1}{f_0} \quad \Rightarrow \quad T = T_0$$

$$t \in \left[-T_0/2, T_0/2\right]$$

$$x(t) = A^2 \sin^2(2\pi f_0 t) = A^2 \left[\frac{1}{2} - \frac{1}{2} \cos(2\pi 2f_0 t) \right]$$

$$x(t) = \frac{A^2}{2} - \frac{A^2}{2} \cos(4\pi f_0 t)$$

Por serie de Fourier $x(t) = a_0 + \sum_{n=1}^N a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$

Serie exponencial $x(t) = \sum_{n=-N}^N C_n e^{jn\omega_0 t} \quad C_n = \frac{a_n - jb_n}{2}$

$$\frac{A^2}{2} - \frac{A^2}{2} \cos(2\omega_0 t) = a_0 + A_2 \cos(2\omega_0 t)$$

Si $\omega_0 = 2\pi f_0$ y $\omega_n = n\omega_0$

$$\omega_n = 2\omega_0 \Rightarrow n\omega_0 = 2\omega_0 \Rightarrow n = 2$$

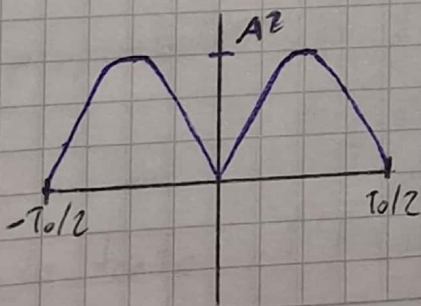
Por lo tanto

$$a_0 = c_0 = \frac{A^2}{2} \quad a_2 = -\frac{A^2}{2} \Rightarrow c_2 = c_{-2} = -\frac{A^2}{4}$$

Por lo tanto $c_n = 0 \quad \forall n \neq \{0, -2, 2\}$

$$a_n = 0 \quad \forall n \neq \{0, 2\} \quad b_n = 0$$

Demostración



Señal par $\Rightarrow x(t) = x(-t)$

$$b_n = 0$$

$$a_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} x(t) \cos(n\omega_0 t) dt = \frac{4}{T_0} \int_0^{T_0/2} A^2 \sin^2(2\pi f_0 t) \cos(n\omega_0 t) dt$$

$$= \frac{4A^2}{T_0} \int_0^{T_0/2} \left[\frac{1}{2} - \frac{1}{2} \cos(2\omega_0 t) \right] \cos(n\omega_0 t) dt$$

$$= \frac{4A^2}{T_0} \left[\frac{1}{2} \int_0^{T_0/2} \cos(n\omega_0 t) dt - \frac{1}{2} \int_0^{T_0/2} \cos(2\omega_0 t) \cos(n\omega_0 t) dt \right]$$

Se sabe que $\cos(2\omega_0 t) \cos(n\omega_0 t) = \frac{1}{2} [\cos(2\omega_0 t + n\omega_0 t) + \cos(2\omega_0 t - n\omega_0 t)]$

Por lo tanto

$$a_n = \frac{2A^2}{T_0} \sin(n\omega_0 t) n\omega_0 - \frac{2A^2}{2T_0} \sin((2+n)\omega_0 t) (2+n)\omega_0$$

$$- \frac{2A^2}{2T_0} \sin((2-n)\omega_0 t) (2-n)\omega_0 \Big|_0^{T_0/2}$$

$$a_n = \frac{2A^2}{T_0} \sin\left(n \frac{2\pi T_0}{T_0} \frac{T_0}{2}\right) n \frac{2\pi}{T_0} - \frac{2A^2}{2T_0} \sin\left((2+n) \frac{2\pi T_0}{T_0} \frac{T_0}{2}\right) (2+n) \frac{2\pi}{T_0}$$

$$- \frac{2A^2}{2T_0} \sin\left((2-n) \frac{2\pi T_0}{T_0} \frac{T_0}{2}\right) (2-n) \frac{2\pi}{T_0}$$

$$a_n = \frac{A^2}{T_0} \left[4n\pi \sin(n\pi) - 2(2+n)\pi \sin((2+n)\pi) - 2(2-n)\pi \sin((2-n)\pi) \right]$$

Como $\sin(n\pi) = \sin((2+n)\pi) = \sin((2-n)\pi) = 0$

Por lo tanto $a_n = 0$ si $n \neq 0, 2$

Si $n=2 \Rightarrow a_2 = \frac{4A^2}{T_0} \left[-\frac{1}{2} \int_0^{T_0/2} \cos^2(2\omega_0 t) dt \right] = \frac{4A^2}{T_0} \left[-\frac{1}{2} \frac{T_0}{4} \right] = -\frac{A^2}{2}$

Si $n=0 \Rightarrow a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A^2 \sin^2(\omega_0 t) dt = \frac{A^2}{T_0} \int_{-T_0/2}^{T_0/2} \sin^2(\omega_0 t) dt$

$$a_0 = \frac{A^2}{T_0} \frac{T_0}{2} = \frac{A^2}{2}$$

$$2. e_r = 1 - \frac{\sum_{n=-N}^N |C_n|^2 P_n}{P_x}$$

$$P_n = \frac{1}{T} \int_{-T/2}^{T/2} |\psi_n(t)|^2 dt = 1$$

$$P_x = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} (A^2 \sin^2(\omega_0 t))^2 dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \left(\frac{1}{2} - \frac{1}{2} \cos(2\omega_0 t) \right)^2 dt$$

$$\left(\frac{1}{2} - \frac{1}{2} \cos(2\omega_0 t) \right)^2 = \frac{1}{4} - \frac{1}{2} \cos(2\omega_0 t) + \frac{1}{4} \cos^2(2\omega_0 t)$$

$$P_x = \frac{1}{T_0} \left[\frac{1}{4} T_0 - \frac{1}{2} \sin(2\omega_0 t) (2\omega_0) + \frac{1}{4} \frac{T_0}{2} \right] \Big|_{-T_0/2}^{T_0/2}$$

$$P_x = \frac{1}{T_0} \left[\frac{1}{4} T_0 + \frac{1}{8} T_0 - \frac{2\pi}{T_0} \sin\left(\frac{4\pi}{T_0} \frac{T_0}{2}\right) + \frac{2\pi}{T_0} \sin\left(\frac{4\pi}{T_0} \left(-\frac{T_0}{2}\right)\right) \right]$$

$$P_x = \frac{1}{T_0} \left[\frac{1}{4} T_0 + \frac{1}{8} T_0 \right] = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$e_r = 1 - \frac{8}{3} \sum_{n=-N}^N |C_n|^2$$

$$3. c(t) = A_c \sin(2\pi F_c t) \quad A_c, F_c \in \mathbb{R} \quad ; \quad m(t) \in \mathbb{R}$$

$$y(t) = \left(1 + \frac{m(t)}{A_c} \right) c(t)$$

$$Y(\omega) = F\{y(t)\} = F\left\{ \left(1 + \frac{m(t)}{A_c} \right) c(t) \right\} = F\{c(t)\} + \frac{1}{A_c} F\{m(t)c(t)\}$$

$$F\{c(t)\} = C(\omega) = F\{A_c \sin(2\pi F_c t)\} = A_c F\left\{ \frac{e^{j2\pi F_c t} - e^{-j2\pi F_c t}}{2j} \right\}$$

Por propiedades $F^{-1}\{X(\omega \pm \omega_0)\} = x(t) e^{\mp j\omega_0 t}$

$$F\{x(t \pm t_0)\} = X(\omega) e^{\pm j\omega t_0}$$

Por lo tanto $F\{x(t) e^{\mp j\omega_0 t}\} = X(\omega \pm \omega_0)$

$$F\{1 \cdot e^{+j\omega_0 t}\} \quad x(t) = 1 \quad X(\omega) = 2\pi \delta(\omega)$$

$$\delta(t) \quad D(t) = 1$$

$$C(\omega) = \frac{A_c}{2j} [2\pi \delta(\omega - 2\pi F_c) - 2\pi \delta(\omega + 2\pi F_c)]$$

$$m(t)c(t) = m(t) A_c \sin(2\pi F_c t)$$

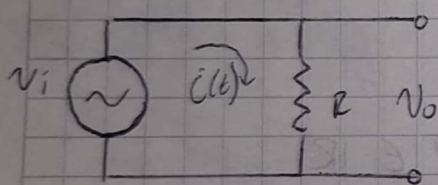
$$F\{m(t)c(t)\} = \frac{A_c}{2j} [F\{m(t)e^{j2\pi F_c t}\} - F\{m(t)e^{-j2\pi F_c t}\}]$$

$$= \frac{A_c}{2j} [M(\omega - 2\pi F_c) - M(\omega + 2\pi F_c)]$$

Por lo tanto:

$$Y(\omega) = \frac{A_c \pi}{j} [\delta(\omega - 2\pi F_c) - \delta(\omega + 2\pi F_c)] + \frac{1}{2j} [M(\omega - 2\pi F_c) - M(\omega + 2\pi F_c)]$$

4. Carga netamente resistiva

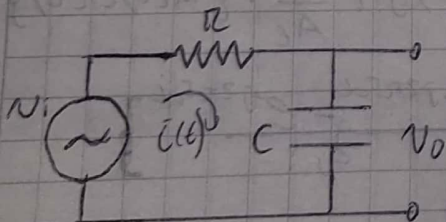


$$i(t) = i_R(t) = \frac{v_R(t)}{R}$$

$$v_i(t) = v_o(t)$$

$$\frac{V_o(s)}{V_i(s)} = 1$$

Carga RC



$$v_i(t) = v_R(t) + v_C(t) \quad ; \quad v_C(t) = v_o(t)$$

$$v_R(t) = R i_R(t) \quad i_R(t) = C \frac{dv_C(t)}{dt}$$

$$V_i(s) = V_R(s) + V_C(s)$$

$$V_i(s) = RCs V_C(s) + V_C(s)$$

$$\frac{V_C(s)}{V_i(s)} = \frac{1}{RCs + 1}$$